MUTASPACE CYBER LEARNING

Information Technology Security

Course Guide and Assignment Checklist Hands-on learning aligned to CompTIA Security+ SY0-701

Interactive course hub for online and in-person sections
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CANVAS IS THE COURSE RECORD	Use Canvas for section announcements, schedules, grades, instructions, feedback, and official assignment submissions.

COURSE OVERVIEW

Learn security by doing the work.

This course is organized as a sequence of realistic security operations. Students investigate, configure, assess, document, and communicate instead of treating Security+ as a vocabulary-only course.

COURSE DESCRIPTION	Instruction in security for network hardware, software, and data, including physical security; backup procedures; relevant tools; encryption; and protection from viruses.

LEARNING OUTCOMES

By the end of the course, students should be able to:

- Classify technical, managerial, operational, and physical security controls.
- Analyze threats, vulnerabilities, indicators, attack methods, and mitigation options.
- Design and explain secure network, system, identity, cloud, and data architectures.
- Harden endpoints, networks, accounts, and services without disrupting required operations.
- Collect and interpret logs, packet captures, alerts, scan results, and other evidence.
- Apply incident response, backup, recovery, and business continuity practices.
- Evaluate risk, governance, policy, vendor, compliance, and security awareness needs.
- Document hands-on projects in a public, sanitized GitHub Pages portfolio.

COURSE RHYTHM

Every operation follows a repeatable workflow.

1. Mission briefing

Review the scenario, authorized scope, success criteria, and evidence requirements.

2. Just-in-time lesson

Learn the concepts needed to begin the security task without receiving the complete solution.

3. Hands-on operation

Configure, test, investigate, analyze, or document within the approved lab environment.

4. Security debrief

Explain the evidence, decision, limitation, and Security+ concept behind the work.

COURSE SYSTEMS

Use each platform for its intended purpose.

PLATFORM	PURPOSE	IMPORTANT NOTE
Canvas	Announcements, schedules, instructions, grades, feedback, and official submissions.	Canvas is the official course record.
Course website	Missions, resources, interactive practice, course guide, and browser-saved progress tools.	Website progress is local and does not submit work.
GitHub and GitHub Pages	Public portfolio, project documentation, README files, and demonstrations of skill.	Sanitize all content before publishing.
Instructor-approved lab	Virtual machines, sample data, logs, packet captures, and authorized practice systems.	Never test real systems without explicit authorization.

COMMUNICATION

Ask for help before a problem becomes a barrier.

For course information or assistance, email **chidinma.agwu@hccs.edu** or send Professor Chidinma Agwu a message through Canvas. Follow the communication expectations posted in your Canvas section.

TECHNOLOGY AND MATERIALS

Prepare the tools needed for hands-on learning.

- A modern web browser and reliable access to Canvas.
- A GitHub account and a public GitHub Pages portfolio repository.
- Access to instructor-approved virtual machines, cloud labs, or provided evidence files.
- A text editor, PDF reader, spreadsheet application, and screenshot tool.
- Optional Security+ books, videos, courses, and lab platforms listed in the Resource Library.

OPTIONAL PURCHASES

No book should be purchased unless your Canvas section identifies it as required. Confirm that any Security+ resource covers SY0-701 and check library access before buying.

ETHICAL AND SAFE USE

Authorization is required for every security activity.

- Use only systems, accounts, networks, files, and targets specifically approved for the activity.
- Do not scan, exploit, disrupt, intercept, or test a real organization network or another person device.
- Do not use course techniques to bypass access controls, obtain credentials, or invade privacy.
- Preserve evidence and document actions accurately. Never fabricate results or alter evidence to improve a report.
- Stop the activity and contact the instructor if the scope is unclear or unexpected behavior occurs.
- Never publish passwords, tokens, private keys, grades, Canvas screenshots, answer keys, real personal information, or restricted lab files.
- Use fictional, sample, or sanitized information in public GitHub documentation.

ASSESSMENT STRUCTURE

Grades reward technical work, evidence, reasoning, and communication.

CATEGORY	GUIDE WEIGHT	WHAT IT MEASURES
Weekly security missions	35%	Technical work, problem-solving, evidence collection, and conclusions.
GitHub security portfolio	20%	Professional documentation, reflection, sanitized evidence, and growth over time.
Security+ knowledge checks	15%	Terminology, concepts, scenario reasoning, and certification readiness.
Midterm practical	10%	Independent completion of an applied security scenario.
Final capstone	20%	Integrated assessment, remediation, incident response, documentation, and briefing.

Canvas contains the official grading details for each course section. The guide weights above describe the planned course structure and may be adjusted in Canvas.

MISSION RUBRIC

Use a consistent standard for hands-on submissions.

AREA	POINTS
Technical task completed correctly	6
Evidence provided and labeled	4
Security reasoning and explanation	4
Documentation and professional communication	3
Safe and ethical lab practices	2
Reflection or recommended improvement	1

GITHUB PORTFOLIO

Build a public record of what you can do.

Every student will create a GitHub account and publish a GitHub Pages portfolio. The portfolio is used for project documentation and professional evidence. Official assignment files and grades remain in Canvas.

- Use a professional username and a public repository suitable for a portfolio.
- Create a clear home page, About section, skills section, project collection, and contact method.
- Document the authorized scope, tools, process, evidence, findings, recommendations, Security+ connection, and reflection for each selected project.
- Write for a public audience. Explain decisions instead of posting unexplained screenshots.
- Add useful alternative text and captions so visual evidence is understandable and accessible.
- Review every page for sensitive data before publishing or sharing the URL.
- Submit the published portfolio address through Canvas when requested.

PUBLIC PORTFOLIO RULE When uncertain, remove or replace the detail. A strong portfolio proves the skill without exposing credentials, private data, restricted files, or an answer key.

PORTFOLIO PAGE FRAMEWORK

Use the same structure for every documented mission.

SECTION	WHAT TO INCLUDE
Context and scope	The approved scenario, systems or data involved, and the boundaries of the activity.
Process and evidence	Tools, steps, screenshots, logs, diagrams, and results that support the work.
Findings and decisions	What the evidence means, the risk or impact, and the control or response selected.
Security+ connection and reflection	Related objective areas, what you learned, and the next improvement.

STUDENT SUPPORT

Use the course systems and contact options early.

- Check Canvas for the instructions that apply to your section.
- Use the course Resource Library for templates, videos, books, tools, practice platforms, and GitHub help.
- Email Professor Chidinma Agwu at chidinma.agwu@hccs.edu or send a Canvas message for course information or assistance.
- Report technical access problems as soon as they affect your ability to complete an authorized activity.

MISSION PATH

Sixteen operations connect technical work to Security+ objectives.

MISSION	TITLE	PHASE	DOMAIN	OBJECTIVES	PORTFOLIO ARTIFACT
01	Security Team Onboarding	Discover	General concepts	1.1, 1.2, 5.1	Baseline assessment summary
02	Asset Discovery	Discover	Architecture	1.1, 3.3, 4.2	Asset inventory case study
03	Network Redesign	Discover	Architecture	1.2, 3.1, 3.2	Network redesign diagram
04	Identity and Access Audit	Protect	Operations	1.2, 4.6, 5.1	Identity and access case study
05	Protect Sensitive Data	Protect	Architecture	1.4, 3.3	Cryptography lab documentation
06	Social Engineering Defense	Protect	Threats	2.1, 2.2, 5.6	Security awareness campaign
07	Suspicious Activity Investigation	Detect	Threats	2.3, 2.4, 4.9	Threat investigation write-up
08	Vulnerability Assessment	Protect	Threats	2.3, 2.5, 4.3, 5.5	Vulnerability management case study
09	System Hardening	Protect	Operations	1.3, 2.5, 4.1, 4.5	Before-and-after hardening report
10	Security Monitoring	Detect	Operations	4.4, 4.7, 4.9	Detection engineering mini project
11	Network Defense	Detect	Architecture	3.2, 4.4, 4.5	Packet analysis and defense report
12	Incident Response	Respond	Operations	4.8, 4.9	Incident response case study
13	Backup and Recovery	Respond	Architecture	3.4, 5.1, 5.2	Backup validation report
14	Risk and Governance	Lead	Program management	5.1 through 5.5	Risk advisory brief
15	Capstone Security Assessment	Lead	Integrated	Integrated review	Capstone technical report
16	Incident Injection and Executive Defense	Lead	Integrated	Integrated review	Final executive presentation

ASSIGNMENT CHECKLIST

Track each stage separately and verify Canvas before submitting.

ASSIGNMENT	STARTED	COMPLETE	CANVAS	PORTFOLIO	NOTES
Course Orientation	[ ]	[ ]	[ ]	N/A	
GitHub Account Setup	[ ]	[ ]	[ ]	[ ]	
GitHub Pages Portfolio Setup	[ ]	[ ]	[ ]	[ ]	
Mission 01: Security Team Onboarding	[ ]	[ ]	[ ]	[ ]	
Mission 02: Asset Discovery	[ ]	[ ]	[ ]	[ ]	
Mission 03: Network Redesign	[ ]	[ ]	[ ]	[ ]	
Mission 04: Identity and Access Audit	[ ]	[ ]	[ ]	[ ]	
Mission 05: Protect Sensitive Data	[ ]	[ ]	[ ]	[ ]	
Mission 06: Social Engineering Defense	[ ]	[ ]	[ ]	[ ]	
Mission 07: Suspicious Activity Investigation	[ ]	[ ]	[ ]	[ ]	
Mission 08: Vulnerability Assessment	[ ]	[ ]	[ ]	[ ]	
Mission 09: System Hardening	[ ]	[ ]	[ ]	[ ]	
Mission 10: Security Monitoring	[ ]	[ ]	[ ]	[ ]	
Mission 11: Network Defense	[ ]	[ ]	[ ]	[ ]	
Mission 12: Incident Response	[ ]	[ ]	[ ]	[ ]	
Mission 13: Backup and Recovery	[ ]	[ ]	[ ]	[ ]	
Mission 14: Risk and Governance	[ ]	[ ]	[ ]	[ ]	
Mission 15: Capstone Security Assessment	[ ]	[ ]	[ ]	[ ]	
Mission 16: Incident Injection and Executive Defense	[ ]	[ ]	[ ]	[ ]	
Midterm Practical	[ ]	[ ]	[ ]	N/A	
Capstone Assessment Package	[ ]	[ ]	[ ]	[ ]	
Final GitHub Portfolio Review	[ ]	[ ]	[ ]	[ ]	

KEEP THE RECORDS SEPARATE

Starting the work, completing the work, submitting it in Canvas, and publishing a sanitized portfolio entry are four different actions.